Distinction Observable Theory. Part I: Foundations and Machine

Author: Igor Zhuk
Affiliation: Independent (Krasnodar, Russia)
Date: March 2026
Series: Distinction Observable Theory (DOT), Part I of III
arXiv subject: Mathematical Physics (math-ph); High Energy Physics — Theory (hep-th)

> "The unmanifest [Prakriti, primal matter] is endowed with the three gunas… The gunas, by their nature, are pleasure [Sattva], pain [Rajas], and indifference [Tamas]. Their purpose is to illuminate, to activate, and to restrict. They suppress each other, support each other, produce each other, and act together. Sattva (light) is considered light and illuminating; Rajas (passion) is stimulating and mobile; Tamas (darkness) is heavy and concealing. They work together for a single purpose, like a lamp [where oil, wick, and flame act in concert].
>
> …
>
> Purusha [pure consciousness] is merely a witness to what occurs. It is endowed with absolute isolation, complete indifference, the capacity to contemplate, and absolute non-action.
>
> …
>
> For the sake of Purusha's contemplation of Prakriti, and for its ultimate liberation, their union occurs. It is like the union of the lame and the blind. From this union, creation arises."
> — *Sankhya Karika, 11–21*

Abstract

The Distinction Observable Theory (DOT; i.e. a theory of observable distinctions) is constructed from a single pre-ontological question: *what must be prohibited for distinction to remain possible?* This first volume establishes the two foundational layers of the theory. **Part I (Foundations)** derives, through purely negative logic, the minimal carrier of distinction — the complete tripartite graph $K(2,2,2)$, an octahedron with six poles, twelve edges, and eight faces. Three Borromean prohibitions eliminate all degenerate modes of collapse; their residual permissions generate three orthogonal axes, six directed cycles, and a unique octahedral topology. No spatial, temporal, or material assumptions are invoked. **Part II (Machine)** shows how a single nilpotent boundary operator ∂ ($\partial^2 = 0$), acting on the octahedral carrier, produces the full operational apparatus: twelve primary distinctions, eight contextual modes, optics (the invariant $\gamma = \sqrt{6}/9$), and the observable artifacts — frame, shadow, tail, and object. The resulting "machine of distinction" is the engine from which all subsequent mathematical and physical structures of DOT emanate. The volume uses a three-tier epistemic status system — [D] (derivation), [G1] (first-order structural correspondence), [G2] (second-order hypothesis) — to maintain a strict boundary between proven results and interpretive extensions.

Keywords: Distinction Observable Theory, boundary operator, Borromean topology, octahedral carrier, $K(2,2,2)$ graph, nilpotency, machine of distinction.

This is Part I of a three-part series. Part II [arXiv:XXXX.XXXXX] derives the full mathematical framework from $\partial^2 = 0$. Part III [arXiv:XXXX.XXXXX] performs the physical readout, including the complete particle spectrum and cosmological parameters.

Introduction

This volume opens the Distinction Observable Theory at its first two levels:

- **Part I** builds the *carrier* — the minimal carrier of distinction;
- **Part II** shows how this *carrier* begins to operate as a *machine*.

The text does not begin with objects, space, or matter. It begins with a more rigorous question: **what must be prohibited for distinction to not collapse entirely**. Therefore, the first part is negative in character. It fixes the impermissible modes of collapse, then shows how permissions, axes, cycles, and ultimately the octahedral *carrier* emerge from them.

The second part no longer builds the carrier from scratch. It accepts the *carrier* as assembled and demonstrates how the following arise from it:

- the first acts of distinction;
- operational and contextual carriers;
- optics;
- frame, shadow, tail, and object as observable artifacts of the machine.

This volume should be read as the **foundation of the entire subsequent DOT**, rather than as a standalone philosophical manifesto. All subsequent parts of the corpus rely on this sequence:

1. negative foundation;
2. legalization;
3. architecture;
4. machine;
5. readout.

To avoid blurring the line between rigorous derivation and stronger interpretations, the entire DOT corpus uses three statuses:

- **[D]** — strict derivation;
- **[G1]** — strong structural correspondence;
- **[G2]** — strongest current reading / hypothesis.

In this volume, the primary material is presented in its most rigorous form. The main goal here is not to explain all physics immediately, but to establish one thesis:

The world in DOT begins not with a thing, but with a permissible distinction that is capable of maintaining its own boundary.

Part I: Negative Foundation

> **"Neti, Neti"**
> — *Brihadaranyaka Upanishad, 2.3.6*

0. Initial Discipline: No External Position

DOT begins with a negative requirement:

There is no external view of the machine of distinction.

This implies four things immediately:

1. We have no right to describe the system "from the outside," as if a neutral observer already existed who is not included in the distinction.
2. Any record is already a local projection. It does not reveal the entire system but only fixes one permissible slice.
3. If the level is not specified, meaning is lost. What is a prohibition at one level becomes an axis at another, a reading mode at the next, and an observable artifact at the level above.
4. Any positive assertion must be prepared negatively: first, one must show what is prohibited, and only then — what is permitted.

Therefore, the foundation of DOT cannot begin with the words "an object exists," "form exists," or "space exists."
First, it is necessary to show **what types of distinction collapse are impermissible**.

---

1. Negative Foundation: Three Invariant Prohibitions

Let there exist a minimal **position of distinction**.
This is not yet a subject, an object, a point in space, or a state.
It is simply the minimal context in which it is permissible to say: "distinction takes place."

If such a position is stated at all, it can collapse in only three fundamental ways.
Each mode of destruction is fixed as an absolute prohibition.

1.1. FC^D — Prohibition of Coincidence

Distinction must not collapse into total identity.

If the distinguisher and the distinguished coincide, the distance between them becomes zero, and the distinction vanishes as a fact.
This means coincidence is prohibited not as a physical event, but as a **logical collapse of distance**.

FC^D maintains the very fact of the cut.

1.2. FC^F — Prohibition of Non-Fixability

Distinction has no right to occur in such a way that no trace remains after it.

Even if distance was maintained, the distinction could become ghostly: the position after the act is indistinguishable from the position before it.
In that case, the distinction is not fixed and logically does not exist.

FC^F maintains the very fact of the trace.

1.3. FC^C — Prohibition of External Grounding

Distinction has no right to justify itself by referring to an external meta-context that is not part of the distinction itself.

If the permissibility of a distinction is explained through an external observer level, an infinite regress of support arises: who legalizes this external level, and so on.

FC^C maintains the very fact of the autonomy of distinction.

1.4. Why exactly three prohibitions?

The importance here lies not in historical choice, but in the completeness of the types of collapse.

If distinction is stated, it can be destroyed:

1. through the disappearance of distance;
2. through the disappearance of the trace;
3. through the disappearance of its own support.

These are three fundamentally different failures:

- coincidence;
- tracelessness;
- external dependence.

They are not derived from each other and are not variants of the same thing.

1.5. Borromean Connectivity of Prohibitions

These three prohibitions do not merely stand side-by-side. They form a minimal Borromean link.

- Remove FC^D — distinction collapses into zero.
- Remove FC^F — distinction dissolves without a trace.
- Remove FC^C — distinction loses its ground and descends into an infinite external support.

None of the three replaces the other two.
Only their joint maintenance makes distinction logically survivable.

At this level, there is not yet:

- space,
- time,
- object,
- axes,
- octahedron.

There is only the minimal stable fabric in which distinction **does not collapse**.

1.6. Formalization

For each position of distinction x, three Boolean predicates are introduced:

- $D(x) = 1$ means: distance is lost (prohibition of coincidence is collapsed);
- $F(x) = 1$ means: trace is lost (prohibition of non-fixability is collapsed);
- $C(x) = 1$ means: autonomy is lost (prohibition of external grounding is collapsed).

Then the admissibility predicate reads as follows:

$$Adm(x) \Leftrightarrow \neg D(x) \wedge \neg F(x) \wedge \neg C(x)$$

A position is admissible if and only if none of the three prohibitions is collapsed.
Weakening any conjunct destroys admissibility:

$$\forall i \in \{D, F, C\} : i(x) = 1 \Longrightarrow \neg Adm(x)$$

This is the formal Borromean connectivity.

1.7. Principle of Inheritance

At the subsequent levels, we will need an operator of level transition. For now, it is sufficient to fix only its requirement:

the next level is obliged to inherit all the prohibitions of the previous one.

$$Adm_{n+1}(x) \Leftrightarrow Adm_n(x) \wedge \text{transition}_n(x) \text{ is defined}$$

The structure of the transition operator itself (∂) will be revealed after the introduction of permissions and axes (§2–§3), once sufficient context for it has appeared.

---

2. Legalization: How the First Positive Layer Arises from Prohibitions

Three absolute prohibitions hold distinction from collapse. But in themselves, they do not yet build anything.
If all three prohibitions act only as rigid stop-signals, the machine remains "clamped": distinction does not break, but it also does not gain a local degree of freedom.

The next step is needed:

- prohibitions must not disappear, but be **redistributed by load**;
- one prohibition becomes the limit;
- the other two form a working corridor.

This is exactly how the layer of **permissions** emerges.

2.1. Two Statuses of Prohibition

At the level of legalization, each prohibition can be in one of two statuses.

Active Status (FC^a)

- the prohibition participates in the form-generation of the mode;
- it holds structural tension;

- it defines the shape of the permissible corridor.

Limit Status (FC^l)

- the prohibition does not participate in building the mode from within;
- it stands as a guardian of the limit;
- it prevents the mode from collapsing into an impermissible position.

Important:

- these are not "two different prohibitions";
- these are two modes of operation for the same prohibition;
- no prohibition can be turned off completely.

2.2. The Permission Formula

A permission reads as follows:

$R \equiv (FC^a \circ FC^a; FC^l)$

Meaning of the formula:

- two prohibitions work as an active corridor;
- the third holds the limit;
- what arises is not the cancellation of a restriction, but a **legal degree of freedom** within it.

Since there are exactly three prohibitions, there are exactly three ways to choose the limit invariant.
This means that instead of an arbitrary number of modes, exactly **three canonical permissions** arise.

2.3. Combinatorics: Why exactly 6 permissible modes?

Each prohibition has two possible statuses: active (a) and limit (l).
Three prohibitions yield a space of $2^3 = 8$ formal configurations of load distribution.

The two extreme configurations are impermissible:

- (a, a, a) — all three prohibitions are active, none stands at the limit $\rightarrow$ the working corridor is unbounded, distinction is not maintained;
- (l, l, l) — all three prohibitions are in limit maintenance, none forms the fabric of the mode $\rightarrow$ the working corridor is absent.

Exactly **6 permissible configurations** remain, for example:

(a, a, l), (a, l, a), (l, a, a) and their polar pairs.

They are grouped into three canonical permissions — by the choice of the limit prohibition.
This result is not postulated but obtained by excluding the extremes from the status space.

2.4. The Three Permissions

R^D

If FC^D becomes the limit, the machine primarily holds the boundary from collapsing.
This is the mode of:

- maintenance,
- density,
- opacity,
- resistance to sticking (merging).

R^F

If FC^F becomes the limit, the machine primarily holds clarity and the trace.
This is the mode of:

- readability,
- manifestability,
- structural luminosity,
- appearance.

R^C

If FC^C becomes the limit, the machine primarily holds internal autonomous mobility.
This is the mode of:

- excitation,
- transfer,
- actualization,
- internal triggering.

2.5. Properties of the set $\{R^D, R^F, R^C\}$

The three permissions inherit Borromean connectivity from the prohibitions:

- no permission is derivable from another;
- none exists in isolation;
- removing the corresponding prohibition destroys the permission;
- permissions **do not have** orientation, sign, or direction — they are not yet axes.

2.6. Limits of Applicability for §2

At the level of legalization, the following are **not yet permitted**:

- sign $+/-$ (no polarity);
- movement or transition between permissions;
- combining permissions as coordinates;
- any forms of geometry or directionality.

Permissions are not yet directions, but only **survivability configurations**.

§2 fixes what survives. §3 will show how the survivor acquires structure.

---

3. Stabilization: The Birth of Axes, Poles, and Limits

§3 is a **transitional** section. It does not introduce objects, quantities, spaces, or dynamics.

It fixes the moment when the permitted modes of distinction (R) cease to be pure possibilities and acquire **stability of form**.

> **§3 answers the question:** Under what conditions does a permission cease to be a local mode and become an axis of the architecture?

3.1. The Principle of Stabilization

Stabilization is a mode of prohibition action such that permission R:

- is **reproducible**;
- is **distinguishable** relative to other R;
- **does not collapse** upon repeated application.

Stabilization adds no new content. It fixes the form of the permission's existence.

> **Stabilization is a condition of repeatability, not change.**

3.2. Sectional Limit Maintenance Operation (⊙)

The operation ⊙ is introduced for the first time.

> **⊙ — the operation of limit maintenance.**

It indicates that an element is in the mode of **active presence as a boundary**, but does not participate in forming the fabric of the mode.

Comparison with previously introduced symbols:

Symbol	Meaning
∘	Coexistence of active prohibitions without their reduction to each other
;	Separation of the form-generating part and limit maintenance
⊙	Fixation of the limit status as a structural guardian

⊙ does not link elements or introduce relationships between them. It fixes the impossibility of going beyond the limit.

3.3. Turning a Permission into an Axis

An axis arise if and only if permission R is maintained by the $\odot$ operation as a reproducible mode.

Formal fixation:

$Axis_D \equiv \odot R^D$
$Axis_F \equiv \odot R^F$
$Axis_C \equiv \odot R^C$

This is **not an identification**, but a fixation of status.

An axis is a stabilized permission.

The designations $Axis_D$, $Axis_F$, $Axis_C$ are introduced intentionally to avoid confusing the axes with the prohibitions $FC^D/FC^F/FC^C$ from §1. Prohibition and axis are different levels: prohibition is an impossibility, axis is a stabilized possibility generated by that prohibition.

3.4. Introduction of Polarity ($\pm$)

Polarity appears **exclusively** at the level of the axis. Before §3, the signs $+$ and $-$ do not exist.

Polarity is the orientation of the axis relative to the limit prohibition that holds its boundary.

For each axis:

Sign	Direction
-	Toward the limit prohibition (return to invariant)
+	Away from the prohibition (realization of the mode)

Important:

- $+$ and $-$ **are not** values, states, or evaluations;
- $+$ and $-$ **are not** symmetrical;
- **polarity is the trace of the prohibition in a stabilized form.**

Note on notation. In the subsequent text, instead of the signs $+$ and $-$, we use the literal statuses a (active — realization) and l (limit — boundary). This is intentional: signs $\pm$ carry a mathematical and everyday semantic history ("plus = good," "minus = absence") which is not only unnecessary here but harmful. The a/l status points exactly to what is meant: active participation in form-generation or limit maintenance. The $\pm$ signs are preserved only in formal records of poles (D^+/D^-) as geometric markers of axis orientation.

3.5. Six Poles

For each axis, two limit projections are fixed:

Axis	Poles
$Axis_D$	D^-/D^+
$Axis_F$	F^-/F^+
$Axis_C$	C^-/C^+

Poles:

- **are not** points;
- **are** modes of approaching the limit.

In total, **six poles** emerge — the first architectural sextet of the theory.

3.6. Borromean Linkage of Axes

Axes $Axis_D, Axis_F, Axis_C$ form a **second-level Borromean linkage**.

Removing any axis:

- destroys the orientation of the others;
- makes polarity meaningless;
- returns permissions to an unstable mode.

Architecture arises not from a single axis, but only from their joint stabilization.

3.7. Threshold of the Architecture

From the moment of axis stabilization:

- **geometric articulation** is permissible;
- **introduction of architectural bodies** is permissible;
- **description of cycles** is permissible.

Configurations (a, a, a) and (l, l, l) are not yet field modes. They are fixed only as **boundary nodes** of the architecture.

3.8. Summary of §3

§3 fixes:

- the $\odot$ operation (limit maintenance);
- the moment of **axis emergence** as stabilized permissions;
- the **origin of signs** $\pm$ from the axis orientation relative to the limit prohibition;
- the **Borromean connectivity** of the axis architecture.

No dynamics, field, or process has been introduced.

4. Cycles: Minimal Closed Forms of Distinction

§4 performs an intermediary function. It transfers the axes from the status of maintained directions to the status of **projectionally observable configurations** and introduces **cycles** as the minimal closed forms of distinction.

> **§4 does not introduce the octahedral body as a whole. It works only with two-dimensional projections of the architecture.**

4.1. The Projectional Principle

Any axis $Axis_D$, $Axis_F$, or $Axis_C$:

- is **not observable** as a whole;
- does **not exist** as a linear object;
- **manifests** only through a joint projection with another axis.

This is a direct consequence of the triadic law (§2): out of three elements, two form a working corridor, while one forms a limit.

Consequently, any observable configuration of axes has the form of a pair: **two axes are active, and the third stands in the limit mode $\odot$ and does not appear in an explicit form.**

4.2. Cycle as a Minimal Closed Form

> **A cycle is a closed projectional configuration of the form:**

$(Axis_X \pm, Axis_Y \mp; \odot\ Axis_Z)$

Where:

- $Axis_X$ and $Axis_Y$ are active axes;
- findings $\pm$ are the limit projections of these axes;
- $Axis_Z$ is the third axis, maintained in the limit mode ($\odot$).

A cycle is not a dynamic, not a process, and not an object.

> **A cycle is the minimal permissible closedness of distinction in projection.**

4.3. Why is the Cycle Closed?

The closedness of the cycle is **not geometric, but based on prohibitions**.

The cycle closes because:

- exceeding the limit of each pole is **prohibited** (FC^D);
- collapsing into the third axis is **prohibited** (FC^C — the axis is in the $\odot$ mode);
- traceless bypass is **prohibited** (FC^F).

Therefore, the only permissible trajectory is a return within a pair of poles. This is the closing.

A cycle is maintained not by a connection of elements, but by a system of prohibitions.

4.4. Three Fundamental Cycles

In total, there are three such cycles — according to the number of ways to choose the $\odot$-axis:

1. $C_D \equiv (Axis_F \pm, Axis_C \mp; \odot\, Axis_D)$ — axis D at the limit
2. $C_F \equiv (Axis_D \pm, Axis_C \mp; \odot\, Axis_F)$ — axis F at the limit
3. $C_C \equiv (Axis_D \pm, Axis_F \mp; \odot\, Axis_C)$ — axis C at the limit

4.5. Borromean Character of Cycles

- Each cycle is **permissible** by itself;
- The removal of any one cycle **destroys** the connectivity of the architecture;
- No single cycle **maintains** the architecture alone.

Cycles form a Borromean linkage of the projectional level: none of the three contains the other two.

4.6. Orientation of Cycles

Each cycle has an orientation: $(+, -)$ or $(-, +)$.

Orientation:

- sets the **direction of projectional tension**;
- **does not set** time;
- **does not set** movement.

This is pre-dynamic asymmetry.

4.7. Prohibited Cycles

Cycles of the form $(+, +; \odot\, C)$ and $(-, -; \odot\, C)$ are impermissible.

Configuration	Why is it prohibited?
$(+, +)$	Requires global coherence — total realization
$(-, -)$	Leads to collapse into the invariant — total return

These configurations are not cycles. They are fixed as limit prohibited forms and will be interpreted in §5 as prohibited nodes (a, a, a) and (l, l, l).

4.8. Status of the Third Axis ($\odot$)

The third axis in a cycle:

- **does not participate** in the projection;
- **does not disappear**;
- **is not active**.

It maintains the boundary of admissibility and prevents collapse.

> **Without ⊙, a cycle either disintegrates or collapses.**

---

5. Faces and the Body of Projections

§5 introduces full configurations of axes (faces) **exclusively as a consequence** of the previously fixed structures (§1–§4).

5.0. Fundamental Constraints

- The body is **not introduced** as an object;
- The body is **not a geometric figure** initially;
- The body is **not a space** of abstract states;
- The body is **not a model** of the world.

> **The body of projections is a canonical map of the permissible configurations of distinction arising from the Borromean linkage of the axes.**

5.1. Full Configuration (Face) as an Element of Architecture

If a pole (§3.5) is the fixation of only one selected axis, and a cycle (§4.2) is the projection of two axes while maintaining the third, then a full configuration arises when selecting one status from each of the three Borromean axes.

> **A full configuration (face) is a minimal contextual linkage of three axes, in which:**
> - each axis has its local status; - a volumetric, complete statement arises; - **the removal of any single axis** destroys the configuration.

Geometrically, such a triplet defines not a vertex, but the choice of one pole on each of the three axes. Spatially, this forms a triangular area — an octant of space or a **Face**.

5.2. Origin of Faces from Axes

Each axis can be read in an active (a) or limit (l) status. The combination of statuses of all three axes gives a **full configuration (context-cell)**.

Using the notation from §2 and §3: - a — active status (realization, $+$ sign on the axis); - l — limit status (return to invariant, $-$ sign on the axis).

There are $2^3 = 8$ possible triplets:

```
(a,a,a)  (a,a,l)  (a,l,a)  (l,a,a)
(l,l,l)  (l,l,a)  (l,a,l)  (a,l,l)
```

These are the 8 faces.

5.3. Prohibited (Blind) Faces: (a,a,a) and (l,l,l)

Face (a, a, a) — Total Realization
- All three axes are in the active status;
- Tension disappears (no limit maintenance);
- The structure loses its hidden complement. Beyond the limit.

Face (l, l, l) — Total Return
- All three axes are in the limit status;
- Distinction collapses in the background;
- Halt of any manifestation.

> **(a, a, a) and (l, l, l) are prohibited as active contexts.**
> **Geometrically, this means that of the 8 faces of an octahedron, two strictly opposite faces are the horizon (blind). One cannot stand on them; they hold the limits.**

5.4. Six Working Faces

Exactly 6 permissible (working) faces remain, forming 3 conjugate pairs. These are configurations containing exactly one a and two l, or exactly one l and two a:

No.	Configuration	Character
1	(a, l, l)	One active
2	(l, a, l)	One active
3	(l, l, a)	One active
4	(l, a, a)	One limit
5	(a, l, a)	One limit
6	(a, a, l)	One limit

Each such face: - **maintains** tension; - **does not collapse**; - **does not lose** distinction.

> **These six dynamic faces form the body of projections.** This forms an absolutely seamless bridge to the Contextual Eight of Part II, of which 6 are working.

5.5. The Body of Projections

The body of projections is not geometry.

> **It is the totality of all permissible (working) contexts of distinction.**

The body of projections: - sets the map of permissible statements; - determines the limits of logical transition; - serves as a support for everything that unfolds thereafter.

No statement lies outside the body of projections.

5.6. Resolution of Topological Balance: 6 Vertices and 8 Faces

In §3.5, we fixed 6 poles (Vertices), where strictly one axis is active.
Here, in §5, we have constructed $2^3 = 8$ full configurations, of which 6 are working.

The Borromean triplet triumphs: - 3 axes generate 6 vertices (permissible poles). - 3 axes generate 8 faces (contextual triplets). - Of the 8 faces, exactly 6 are working (dynamic), and 2 are absolutely blind horizons.

The mathematics converges perfectly. The octahedron is verified without error.

---

6. The Octahedron as a Representational Structure

After the derivation of 6 poles and 8 contextual configurations was completed in §5, the question arises:

What is the minimal connected geometric form in which these elements can coexist without violating any prohibition?

It is important to emphasize: we are looking for a **representation** — a map of permissible positions on which the architecture of distinction acquires geometric readability.

6.1. Requirements for Representation

From §1–§5, the following rigid conditions follow: - the three axes must be orthogonal (irreducible to each other); - each axis has two poles (a and l); - antipodes must be separated (prohibition of coincidence, FC^D); - the central point — where all three axes "intersect" — is not permissible (prohibition of external grounding, FC^C); - the structure must be finite and closed (prohibition of tracelessness, FC^F).

6.2. Why exactly the Octahedron?

The minimal geometric structure satisfying all five conditions is the **octahedron**.

The octahedron is the only regular polyhedron in which: - there are exactly three axes; - each has exactly two antipodal poles; - all poles have the same degree (4); - the center is not a permissible node; - the structure is closed and finite.

The octahedron was not chosen for beauty — it is **necessitated** as the minimal topological representation.

6.3. Why not another figure?

The carrier cannot be a **cube**: a cube has 8 vertices and 4 main diagonals — it is a permissible extended vector graph (dual to the octahedron's faces), but not minimal for the basis of three

primary axes. The carrier cannot be a **tetrahedron**: it does not contain antipodal pairs or orthogonality.

6.4. Vertices, Edges, and Faces

As a canonical representation, the octahedron contains:

Element	Number	What it is
Vertices	6	6 absolute poles (2 per axis from §3.5)
Edges	12	12 transition paths (instantiations of the ∂ operator) between non-antipodal poles
Faces	8	8 contextual configurations (one projection from each axis from §5)

These numbers are linked by Euler's formula: $V - E + F = 2 \to 6 - 12 + 8 = 2$.

Vertices localize (poles). Edges transfer tension (acts). Faces maintain semantic contexts (cells).

Of the eight faces: - 6 are dynamic (working mixed); - 2 are blind horizons (a, a, a) and (l, l, l). They fix absolute limits and do not participate in dynamic traversal.

6.5. Center-Prohibition

There is no geometric center in the architecture — only a **center-prohibition**: a node of impossibility that doesn't belong to any of the axes.

6.6. From Representation to Operator

Up to this point, the octahedron is only a **map**: a static representation of permissible configurations of distinction. Positions and links are marked on the map, but no one traverses it yet.

However, this map has a property that makes it something more than just a representation.

Each pole on the octahedron has exactly **four** non-antipodal neighbors. This means that from any permissible position, exactly four permissible transitions exist. The map has no dead ends, no degree asymmetries, and no privileged positions.

Such an equal-degree structure does not need an external triggering mechanism. If distinction once finds itself in a permissible position, it **cannot fail to have** routes of departure — they are built into the geometry itself.

This transition — from a map to something that *traverses itself* — is the transition from representation to the **operator**. But the operator, its acts, and its routes are the subject of Part II.

6.8. Closing the Foundations

Element	Status
Center	Prohibited
Axes	Defined
Vertices	Limiting
Edges	Connect non-antipodal poles
Faces	Hold admissibility

After §6:

- no new foundations can be introduced;
- the number of axes cannot be changed;
- a center cannot be legally added.

All further work is traversal and unfolding around the prohibited center, not a deepening into it.

> **The octahedron is not a model of the world, but a map of the impossibility of the center, within which all observable dynamics become possible.**

Canonical Derivation Chain

```
§1 Prohibitions (FCᴰ, FCᶠ, FCᶜ)
  → §2 Permissions (R^D, R^F, R^C) — 6 permissible configurations
    → §3 Axes (⊙R), Poles (±), Limits — 6 poles
      → §4 Cycles (2 axes + ⊙ third) — 3 Borromean cycles
        → §5 Nodes (8 → 6 permissible) — body of projections
          → §6 Octahedron — carrier structure (C0)
```

The Carrier is assembled.

At each step of this chain, the same principle operates: **Borromean connectivity** passes from level to level, never once breaking.

Level	What is Borromean-linked
§1	Three prohibitions {FC^{D}, FC^{f}, FC^{c}}
§2	Three permissions {R^D, R^F, R^C}
§3	Three axes {Axis_D, Axis_F, Axis_C}
§4	Three cycles {C_D, C_F, C_C}

> **"That Thou Art [Tat Tvam Asi]."**
> — *Chandogya Upanishad, 6.8.7*

Part II: The Machine

The carrier is now assembled: six poles, twelve edges, eight faces, the prohibited center. This is the minimal topological architecture compatible with the three invariant prohibitions.

But the carrier itself is not yet a machine. It is the frame on which the machine can be launched.

For the carrier to begin to **act**, it is necessary to:

- introduce the first acts (`1|2`, `2|3`);
- show how acts are projected onto edges (operational 12) and faces (contextual 8);
- define the reading mode (optics);
- and only then can one speak of observable artifacts.

All of this is the subject of **Part II: The Machine**.

---

7. Pre-machine Root: The Triad of Primal Bases

Before launching the octahedron as a machine, it is necessary to fix what exactly it projects. The octahedron (§6 of Part I) is already the first full representation, but it does not arise from nowhere.

At the foundation of DOT lies a pre-machine triadic root. The world does not consist of things, but of maintained distinctions. And these distinctions rely on three primal acts:

- **Separate** (establish a boundary)
- **Manifest** (distribute the explicit and the hidden)
- **Actualize** (set in motion)

Individually, they are not maintained: a boundary without manifestation is empty, manifestation without a boundary is blurred, and action without the first two has no carrier. Only their Borromean linkage yields what we call the *machine*.

7.1. The Triad: Boundary / Meaning / Action

This triplet of primal acts can be fixed as three ontological primal bases. These are not yet the poles of the octahedron (there will be six of them). They are that *from which* the octahedron unfolds. To maintain the theory's rigorous law ("nothing is introduced from outside"), it is necessary to explicitly fix: the Triad is the positive semantic translation of the three negative prohibitions from Part I.

1. **Boundary (Form/Boundary)**
   The consequence of FC^D (prohibition of coincidence). That which has no right to merge into identity is logically presented as a Boundary. This is the act of separation that makes "this" and "not-this" possible (in early DOT terminology: **Tamas**).
2. **Meaning/Idea/Trace (Idea/Manifestation)**
   The consequence of FC^F (prohibition of non-fixability). That which has no right to disappear without a trace is presented as the very maintenance of the manifest and the hidden. This is the act of highlighting what has become readable and what has gone into shadow (in early DOT terminology: **Sattva**).

3. **Action (Act/Action)**
   The consequence of FC^C (prohibition of external grounding). The compulsion of distinction to rely only on itself generates self-actualization. This is the transfer, the step, the path along which the system moves from the potential to the unfolded (in early DOT terminology: **Rajas**).

Thus, the "negative foundation of algebra" (Part I) is firmly fused with the "semantics of the machine" (Part II), and the pre-machine triad Tamas/Sattva/Rajas acquires a rigorous structural basis.

7.2. The Principle of Projection

An important epistemic fact of DOT: **we never describe the world from the outside.** We are always already within some projection.

Therefore, the raw *carrier* of distinction (our octahedron with its 6 vertices, 12 edges, and 8 faces) cannot be seen with an "absolute gaze." We will read it through the first and most fundamental projection — the **Machine projection (Machine Family)**, where this primal basis is polarized into 6 local poles.

For this reading to begin, the machine must produce an unfolding of form. This unfolding does not occur instantaneously, but along a rigorous ladder of acts.

---

8. The Ladder of Acts: From Identity to Locality

The machine arises not from a single principle, but from a ladder of transitions: from the zero pre-distinguished level to the first full structure ($1 \to 1|2 \to 2|3 \to 6$).

8.1. The Whole before Distinction: 1

DOT begins not with a world object, but with self-identity (anchor).
1 is the primary invariant of self-consistency.

At this level, form and action are not yet separated. The machine is not yet launched; there is nothing to distinguish.
But without self-identity, it is impossible to draw a boundary and understand what exactly will be maintained in the machine cycle as one and the same.

From this arises the most important pre-linguistic law of DOT:
A part is a mode of access to the whole, rather than a fragment of the whole.

- $1/n$ is the minimal available representation of the whole in a given frame (the way it is forced to appear).
- n is the inverse reading of the same whole as a self-identical form.

At this same minimal numeric slice, a neighboring boundary-law arises:

[=1-1n.]

It yields the simplest numeric readout of any boundary:

[B(n)=(=n-1, =n, =1n).]

Namely:

- the `carrier` is set by the number n;
- the `body` is read as almost complete filling $n - 1$;
- the `debt` is the minimal hidden remainder $1/n$.

Hence:

- $2/3 = 1 - 1/3$ — triadic manifestation with a hidden share;
- $3/4 = 1 - 1/4$ — fourfold readable frame with a witness-slot.

This does not replace machine syntax but provides its simplest numeric slice.

8.2. Establishment of the Boundary (Cut): $1|2$
The first true act of the machine. The unified is sliced by a two-sided boundary.

It: - draws the boundary (`this / not-this`); - establishes **two sides** as mutually defined; - creates an involution (flipping the frame reverses the sides but leaves the boundary unchanged).

$1|2$ makes the world distinguishable (locally readable), introduces two-sidedness, but does not yet yield a directed manifestation of the world. A pair (Inside/Outside) arises.

8.3. Distribution of Manifestation (Split): $2|3$
The second act of the machine. For distinction to become a full-fledged object of reading, a triadic split arises.

It: - gives the first true direction to manifestation; - maintains two shares in focus (manifest) and pushes one into the hidden (hidden share); - makes manifestation inherently incomplete.

Here, a DOT law first appears: **any local readability leaves a hidden share.**
$2|3$ creates asymmetry and shadow, but without the two-sided boundaries of $1|2$, this shadow could not be stably maintained.

8.4. The First Full Register: $2 \times 3 = 6$
As soon as the two-sided boundary operator ($1|2$) is applied to each of the three independent directions of manifestation ($2|3$), the three primal acts are polarized, receiving two ends (poles) each.

This is the combinatorial limit: $2 \times 3 = 6$.
Two sides are applied to the triad of primal bases, yielding the *Machine Six*. This is the first level where a stable machine object appears, rather than just the conditions for its possibility.

---

9. Machine Six: Polarization of Primal Bases

Each of the three primal bases (Boundary/Idea/Action) is polarized by a two-sided cut and turns into a readable pair on the axes of the octahedron ($Axis_D$, $Axis_F$, $Axis_C$ — from Part I).

Thus arises *Machine Six* — the first point at which the raw carrier of distinction can be described as a set of local poles.

9.1. Polarization of the Boundary (Form $\to Axis_D$)

The primal base of Boundary unfolds as an axis of maintenance:

- `Inside` — local maintenance (the form we fix)
- `Outside` — external complement (that from which the form is isolated)

This is the first axis of the machine projection: **Inside / Outside**.

9.2. Polarization of Meaning/Idea (Idea $\to Axis_F$)

The primal base of Idea/Manifestation unfolds as an axis of fixability:

- `Manifest` — revealed (the structure that is read locally)
- `Shadow` — hidden (the semantic and form-generating remainder)

This is the second axis of the machine projection: **Manifest / Shadow**.

9.3. Polarization of Action (Act $\to Axis_C$)

The primal base of Action unfolds as an axis of actualization:

- `State` — maintained action (fixation in a point, potential)
- `Path` — unfolded action (openness to transition, vectorness)

This is the third axis of the machine projection: **State / Path**.

9.4. Machine Six as a Dictionary

Thus emerges the first working basis of the machine:

- `Inside / Outside`
- `Manifest / Shadow`
- `State / Path`

`6` is not six things of the world. It is the **first full translation of DOT into a locally readable language**. It is in this language that the machine will begin to carry out transitions (operational twelve: 12) and close into small worlds (contextual eight: 8). The bridge is laid.

---

10. Operational Twelve: How Structure Begins to Act

As soon as `6` becomes the carrier, the machine automatically unfolds a layer of actions.

10.1. Why 12 and where the ∂ operator is hidden

Each vertex of the octahedron has four permissible non-antipodal neighbors.

This means: - each pole has a fan of 4 permissible transitions; - in total, there would be 6×4 local moves; - but each edge is counted twice.

Hence: $12 = 6 \times 4/2$

This is the first *operational layer*: - $12 = C_1$

It is here that the transition operator ∂, promised in Part I (§1.7), is legalized. Each of the 12 edges is a specific vector ∂, transferring the machine from one status (pole) to another without losing Borromean connectivity. The octahedron of paths is the graph of all permissible acts of ∂.

Here, the structure no longer merely exists but acts as a path-space.

10.2. 12 as a Periodic Table of Actions

The *operational twelve* cannot be read as a flat list of edges.
Each edge connects exactly two axes, meaning the entire twelve is divided into three families of four actions each.

Class A: Boundary ↔ Manifestation

Connections between:

- `Inside / Outside`
- `Manifest / Shadow`

This family is responsible for how a boundary becomes manifest or hidden.

Here reside actions of the type:

- localization of manifestation;
- internal hidden maintenance;
- exposition;
- retreat into the external background.

Class B: Boundary ↔ process

Connections between:

- `Inside / Outside`
- `State / Path`

This family is responsible for how a boundary triggers or restricts the flow.

Here reside:

- occupation;
- launch;
- environmental loading;
- route restriction.

Class C: Manifestation ↔ process

Connections between:

- `Manifest / Shadow`
- `State / Path`

This family is responsible for how the explicit and the hidden control the action.

Here reside:

- objectification;
- expression / transfer;
- latent retention;
- deferred transfer.

Thus, 12 is no longer just a number of edges, but the first systematic table of machine actions.

10.3. Path-length and local fan

The *operational layer* introduces:

- path length;
- local fan;
- transition order;
- and the first internal time of the machine.

Time here is not yet an external clock.
It appears as the order of traversing permissible paths.

---

11. Contextual Eight: How the Path Closes into a Local World

If 12 provides routes, then a layer is needed where the route ceases to be merely a transfer and closes into a local context.

This is exactly what 8 does.

11.1. Why 8

To define a local context, the machine chooses:

- one pole from each of the three axes.

On each axis, there are two poles.
Thus, the number of local three-pole selections equals:

$$2 \times 2 \times 2 = 8$$

These are the eight faces of the octahedron:

$$8 = C_2$$

11.2. What a context-cell is

Each face:

- takes one pole from each axis;
- closes three edges into a local cell;
- forms the first full local context.

Thus, 8 is not just another carrier alongside 6 and 12, but the first layer of:

- local worlds,
- closure-cells,
- self-sufficient small contexts.

11.3. Four pairs of opposite faces

The eight faces are organized as:

- 4 pairs of antipodal triangles.

This means:

- any chosen *context-cell* automatically has a conjugate *shadow-cell*;
- a local *frame* always drags along with it a strictly opposite hidden complement.

It is here that the geometry of frame/shadow first becomes rigid.

11.4. Why 8 is fundamentally important

Without 8, the machine:

- knows how to distinguish,
- knows how to lay paths,
- but does not know how to close a path into a local world.

Namely: - 6 — what is distinguished; - 12 — how it acts; - 8 — where it locally assembles into a closure.

---

12. Optics: How Topology Becomes Locally Readable

Only after 6/12/8 is it permissible to introduce not the topology of the machine itself, but its **reading mode**.

This is optics.

12.1. Syntactic Hygiene

One must strictly distinguish four types of notation:

- `|` — operator act
- `/` — local aperture
- `⟨·⟩` — readout / specific numerical or semantic reading
- `×` and integers — carrier / machine body

Therefore:

- `|2/3|` — operator triadic separation;
- $2/3$ — frame state;
- $\langle 2/3 \rangle$ — specific reading of this frame;
- $2 \times 3 = 6$ — structural body from which this reading arises.

Summary Table:

View	Sign	What it means
Operator	\\\|	act (cut, split)
Frame	/	aperture / visibility state
Reading	$\langle\rangle$	readout / specific value
Carrier	$\times$, integers	carrier / machine body

12.2. Dyadic Aperture $1/2$

This is the minimal form of readability through a pair of sides:

- yes / no;
- figure / background;
- inside / outside.

Here, the very fact of two-sided distinguishability is read.

12.3. Triadic Aperture $2/3$

This is the aperture of manifestation.

It maintains:

- two shares in focus;
- one share in the *hidden share*.

This is the earliest readable form of triadic asymmetry.

12.4. $3/4$ **as a Second-Level Projected Tone**

$3/4$ cannot be read as "the third fraction of the same first row."
It is already a *local aperture* on C_1.

At the vertex of the octahedron:

- degree is 4;
- *local fan* = 4 permissible channels.

But the *triadic law* still requires *hidden support*.
Thus, one of the four channels must become:

- a blind anchor;
- hidden support;
- witness-slot.

Then locally there remains:

$$3 \ \text{ manifest channels} + 1 \ \text{ hidden witness-slot}$$

This is the correct reading of $3/4$.

12.5. 1/4

The hidden quarter here is not external garbage nor simply a "remainder."

It performs the function of:

- witness,
- support,
- folded return,
- hidden local self-support.

Thus, 1/4 is the hidden support for the readability of 3/4.

---

13. Artifacts: What remains after the machine read-out

Only after the assembly of topology and optics does a layer of phenomena arise.

13.1. Frame and the Arrow of Time

`Frame` is a *context-cell* selected by a local aperture as the current world.

A *context-cell* itself is not yet a *frame*. It becomes a *frame* only when optics (§12) selects it as the point of application for the current reading. Before that, all eight cells are equal.

Discovery of the Arrow of Time: It is here that time arises. Time is not an external parameter, but a *sequence of witness-frame switchings*. When optics fixes one *context-cell* as a *frame*, the cell conjugate to it goes into shadow (*shadow*). The next step of the machine is a new selection of a contextual cell. Thus, the arrow of time is directed toward increasing the number of read frame-switchings.

13.2. Shadow

`Shadow` is the conjugate *shadow-cell* opposite to the frame.

The shadow is not an emptiness but a hidden structural complement to the *frame*.

Shadow is not a lost remainder or a projection error. It is a conjugate cell that contains exactly those poles that the *frame* did not select. Without a shadow, a *frame* has no defined boundary: it is the presence of the conjugate complement that makes the frame finite and closed.

13.3. Tail

The tail does not lie at the foundation.
It arises when the partial local *optics* fails to read the *topology* without loss.

It can be read in two compatible ways:

1. **topology/optics debt**
   the local aperture loses part of the full structure;
2. **face-pair tension**
   an uncollapsed tension remains between the selected *context-cell* and its conjugate *shadow-cell*.

The tail is the observable weight of the unremoved shadow. Taking *face-pair tension* into account, this irreducible residual structure can be canonically described as a **low-rank tensor**, where the diagonal acts as *direct response*, and the off-diagonal elements as *resonant tension* between conjugate faces.

The tail is not an error or a random discrepancy. It is structural: it arises every time a local aperture attempts to read the machine, but the completeness of the topology does not fit into the reading frame. The tail is always present as long as the reading remains a projection.

13.4. Object

An object is not a point.
And it is not just a *carrier* of poles.

An object is:

the self-consistent closure of a traversal,
locally consistent with its shadow and its tail.

This is no longer a primitive but the result of a successful closure.

An object is not a point in space and not a ready-made thing. It is a closed path, consistent with its shadow and having a certain tail. An object arises when a traversal returns to the initial position without destroying either the *frame* or the *shadow*.

13.5. Interaction

Interaction does not reside within a single bare edge.
It begins where:

- two *register-bodies* meet;
- their *frame/shadow* geometry overlaps;
- the total tail is redistributed.

That is, interaction lives not on an *isolated edge*, but on a *cross-register overlap*.

---

14. Projections: How One Machine Generates Different Dictionaries

The raw carrier is not reducible to one of the dictionaries. Depending on the chosen reading frame (projection), it generates different families of definitions.

14.1. Machine projection

Polarization of the raw supporting frame into structural roles:

- Boundary → `Inside / Outside`
- Idea → `Manifest / Shadow`
- Action → `State / Path`

This is the internal dictionary of the machine, the first complete language of its description.

14.2. World projection

A semantic reading of the same structure as a phenomenal world:

- `Form / Space` (maintained local / external carrier)
- `Matter / Idea` (manifest body / hidden organizing information)
- `Energy / Time` (intensity of actualization / sequence)

Here, in the place of human "consciousness" stands **Idea** — as the hidden form-generating pole of matter.

14.3. Logic projection

A logical reading of the same machine as an inference space:

- `True / False` (boundary polarization)
- `Explicit / Implicit` (manifestation polarization)
- `Statement / Derivation` (act polarization)

From here follows an important epistemic conclusion:

Any dictionary itself has a tail.
Since Machine projection, World projection, and Logic projection are all optics of the L5 level, our own canonical language for describing DOT is also a projection and, consequently, generates its own irreducible *tail*. By choosing one reading mode (e.g., logical), we algorithmically hide other permissible projections of the same carrier. The recursion is perfect.

---

15. The Canonical Stack L0–L6

Now the whole can be assembled.

L_0 — **Negative Foundation (Part I, §1)**
- FC^D
- FC^F
- FC^C

L_1 — **Legalization (Part I, §2)**
- $R^D/R^F/R^C$
- 6 permissible configurations

L_2 — **Architecture (Part I, §3–§6)**
- Axes, Poles, Cycles, Nodes
- Octahedron = C_0

L_3 — **The Root and the Ladder (Part II, §7–§9)**
- Triad: Boundary, Meaning/Idea, Action
- Ladder: $1 \to 1|2 \to 2|3 \to 6$
- Octahedron is projected into Machine Six

L_4 — **Operational and contextual topology (Part II, §10–§11)**
- $12 = C_1$
- $8 = C_2$
- path-space
- context-cells
- conjugate face-pairs

L_5 — **Optics (Part II, §12)**
- $1/2$
- $2/3$
- $3/4$

L_6 — **Artifacts (Part II, §13)**
- frame
- shadow
- tail
- object
- interaction
- readouts

The order of this stack cannot be rearranged without destroying the logic of derivation.

---

16. What Constitutes the Canon here

Canon
- absence of an external position;
- three invariant prohibitions;
- three permissions;
- stabilization of permissions into axes ($\odot$);
- cycles and Borromean linkage;
- six permissible nodes and the octahedral carrier;
- the pre-machine triad (Boundary, Idea, Action) and the ladder of acts ($1|2, 2|3$);
- Machine Six: `Inside / Outside`, `Manifest / Shadow`, `State / Path`;
- `6`, `12`, `8`;
- the distinction between operator / carrier / optics / artifact;
- $3/4$ as a second-level local projected tone;
- $1/4$ as a hidden witness/support slot;
- object as a self-consistent closure;
- tail as topology/optics debt and face-pair tension.

Strong but not yet final readings
- deeper tensor-like formalization of the tail;
- further higher-level carriers;

- rigid linking with external traditions;
- full emanation of mathematics and physics from this framework.

Borromean chain across levels
Borromean connectivity is repeated at each level of the stack:

Level	What is Borromean-linked
§1 (Part I)	Three prohibitions $\{FC^D, FC^F, FC^C\}$
§2 (Part I)	Three permissions $\{R^D, R^F, R^C\}$
§3 (Part I)	Three axes $\{Axis_D, Axis_F, Axis_C\}$
§4 (Part I)	Three cycles $\{C_D, C_F, C_C\}$
§11 (Part II)	Three working conjugate pairs of faces

Borromean connectivity is not a property of one level, but an **invariant of the entire theory**.

---

17. Transition to Part III: Emanation of Mathematics

This foundation has prepared the ground for pure mathematics (Syntax of Distinction).
The *Contextual Eight* (L4 layer) act as the exact topological carrier for the mathematical constants and operators that will appear in Part III: - $\boldsymbol{\pi}$ emanates as resonance (the necessary coordination of face-pair tension); - $\boldsymbol{i}$ emanates as the horizon of invisibility (the kernel of the transition operator, $\ker(\partial)$); - $\boldsymbol{\nabla}$ (nabla) emanates as the full unfolding of the stress gradient along all available paths 12.

Mathematics does not precede distinction. Numbers, geometry, and constants themselves emanate from the configurations of the octahedral framework.

---

18. The Brief Formula of the Machine

If the entire foundation is condensed into the shortest logic, the result is:

1. no external position;
2. distinction survives only with the joint maintenance of three prohibitions;
3. this gives rise to three legalization modes;
4. the modes stabilize into axes ($\odot$);
5. axes generate three Borromean cycles;
6. cycles combine into six permissible nodes (octahedral carrier);
7. at the base of the projection lie Boundary (Tamas), Meaning (Sattva), and Action (Rajas);
8. they are polarized through the ladder of acts ($1|2, 2|3$) into *Machine Six*;
9. the carrier unfolds into 12 paths (∂) and 8 contextual cells;
10. optics reads the topology partially ($1/n$ as access to the whole);
11. **the arrow of time manifests as a sequence of witness-frame switchings;**
12. shadow and tail (low-rank tensor) arise as the structural debt of this locality;

13. the object manifests as a locally self-consistent closure of traversal.

This is exactly how the machine of distinction arises in DOT.

Distinction Observable Theory — Part I: Foundations and Machine
Canonical revision for publication, March 22, 2026